Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		furthest line to the left	1			1		
		(ii)		electron falls from higher energy levels to lower energy levels / to n = 2 (1) the difference between any two energy levels is fixed / energy levels are quantised (1)	2			2		
	(b)	(i)		this is the energy required to remove an electron from an atom of hydrogen / for an electron to go from n = 1 to n = ∞ (1) in the gaseous state (1) [award (1) for correct equation with state symbols but no explanation]	2			2		
		(ii)		E = hf (1) $f = 2.18 \times 10^{-18} / 6.63 \times 10^{-34}$ (1) $f = 3.29 \times 10^{15}$ (1)	1	2		3	3	
	(c)			$V_2 = \dfrac{163 \times 273}{398} = 112$ (1) n hydrazine = 112/22400 = 0.00500 (1) M_r hydrazine = 0.160/0.005 = 32 so molecular formula is N_2H_4 (1) ecf possible credit alternative method using pV = nRT		2	1	3	2	